FORMATIVE ASSESSMENT 1
LAB ACTIVITY

1. Aim

To observe a chemical displacement reaction.

2. Apparatus

- Hydrochloric acid
- Copper (II) sulphate solution
- Two test tubes
- Iron nails
- Sand paper

3. Procedure

1. Take two iron nails and clean them by rubbing with sand paper.
2. Take two test tubes and mark them as A and B.
3. Take about 10ml of copper sulphate solution in each test tube. Record the physical properties of the solution.
4. Dip one iron nail in the copper sulphate solution of test tube A and keep it undisturbed for 20 minutes.
5. Keep the other iron nail and test tube aside.
6. Compare the colours of the solutions in the test tubes.
7. Now take the iron nail from the copper sulphate solution and compare with the other iron nail that has been kept aside.

4. Observations

1. There is a brown coating on the iron nail dipped in the copper sulphate solution. Whereas the iron nail placed in the Petri dish shows the grayish colour of iron.
2. The colour of the solution of copper sulphate in which the iron nail was dipped changes to light greenish, whereas the colour of copper sulphate solution in the other test tube does not change.

5. Analysis

1. The brown coating on the iron nail shows that copper is deposited on the iron nail by displacing iron.
2. The greenish colour of the solution in the test tube shows that Fe^{2+} ions are present in the solution.
3. This shows that iron is more reactive than copper, as iron has displaced Cu^{2+} ions from the copper sulphate solution.

6. Chemical Equation

$$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$$

7. Result

This is a single displacement reaction in which copper has been displaced by iron from the copper sulphate solution.

8. Precautions

1. Clean the iron nails by rubbing with sand paper to remove rust, dust or greasy surface.
2. Avoid touching the copper sulphate solution or the nail dipped in the copper sulphate solution as copper sulphate is poisonous.

